CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

Case 15-2026: A 64-Year-Old Woman with Fatigue, Memory Changes, and Falls

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PRESENTATION OF CASE

Dr. Sumita M. Strander (Neurology): A 64-year-old woman was admitted to this hospital because of fatigue, memory changes, and falls.

The patient had been in her usual state of health until 3 weeks before the current presentation, when fatigue developed during a summer vacation in Maine. She had spent time outdoors near a lake but had had no known tick bites.

One week before the current presentation, the patient had increasing fatigue while driving. She parked the car and called her husband, who took her to the emergency department of another hospital for evaluation.

In the emergency department, the patient reported fatigue and poor sleep. She had no dysuria, chest pain, or abdominal pain. A physical examination and an electrocardiogram were reportedly normal. The blood levels of electrolytes, troponin, thyrotropin, alanine aminotransferase, aspartate aminotransferase, bilirubin, and alkaline phosphatase were normal, as were the complete blood count with differential count and the results of tests of kidney function. Urinalysis showed trace ketones (reference value, negative) and 1+ leukocyte esterase (reference value, negative), and microscopic examination of the urine sediment revealed many squamous cells and 6 white cells per high-power field (reference value, <4). A test for Lyme disease was negative. The results of computed tomography (CT) of the head, performed without the administration of intravenous contrast material, were reportedly normal. The patient was discharged home with plans to follow up with her primary care physician.

The next day, the patient was evaluated in the primary care clinic. In addition to ongoing fatigue, she reported difficulty falling asleep, poor sleep quality, poor memory, and occasional lightheadedness. A diagnosis of anxiety was considered, and treatment with escitalopram and as-needed ramelteon was started. Results of the urinalysis performed at the other hospital were reviewed, and nitrofurantoin was prescribed.

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CME

Two days later, and 4 days before the current presentation, the patient began to have an unsteady gait, and her husband took her to the emergency department of a second hospital for evaluation. She reported feeling "off balance" while walking and having difficulty with concentrating on tasks. She had no headache or change in vision. Her performance on finger-to-nose testing was accurate but slow. She required multiple attempts to correctly touch her left thumb to her right ear and was unable to calculate the value of nine quarters; these findings differed from baseline findings. Her gait was cautious, but she was able to ambulate without support. The remainder of the examination was normal. Results of tests of coagulation were normal. The erythrocyte sedimentation rate was 7 mm per hour (reference range, 0 to 30), and the blood level of C-reactive protein was 1.2 mg per liter (reference range, 0.0 to 10.0). Nucleic acid amplification testing for ehrlichia and anaplasma DNA was negative.

Dr. Samuel C.D. Cartmell: CT of the head (Fig. 1A), performed without the administration of intra-

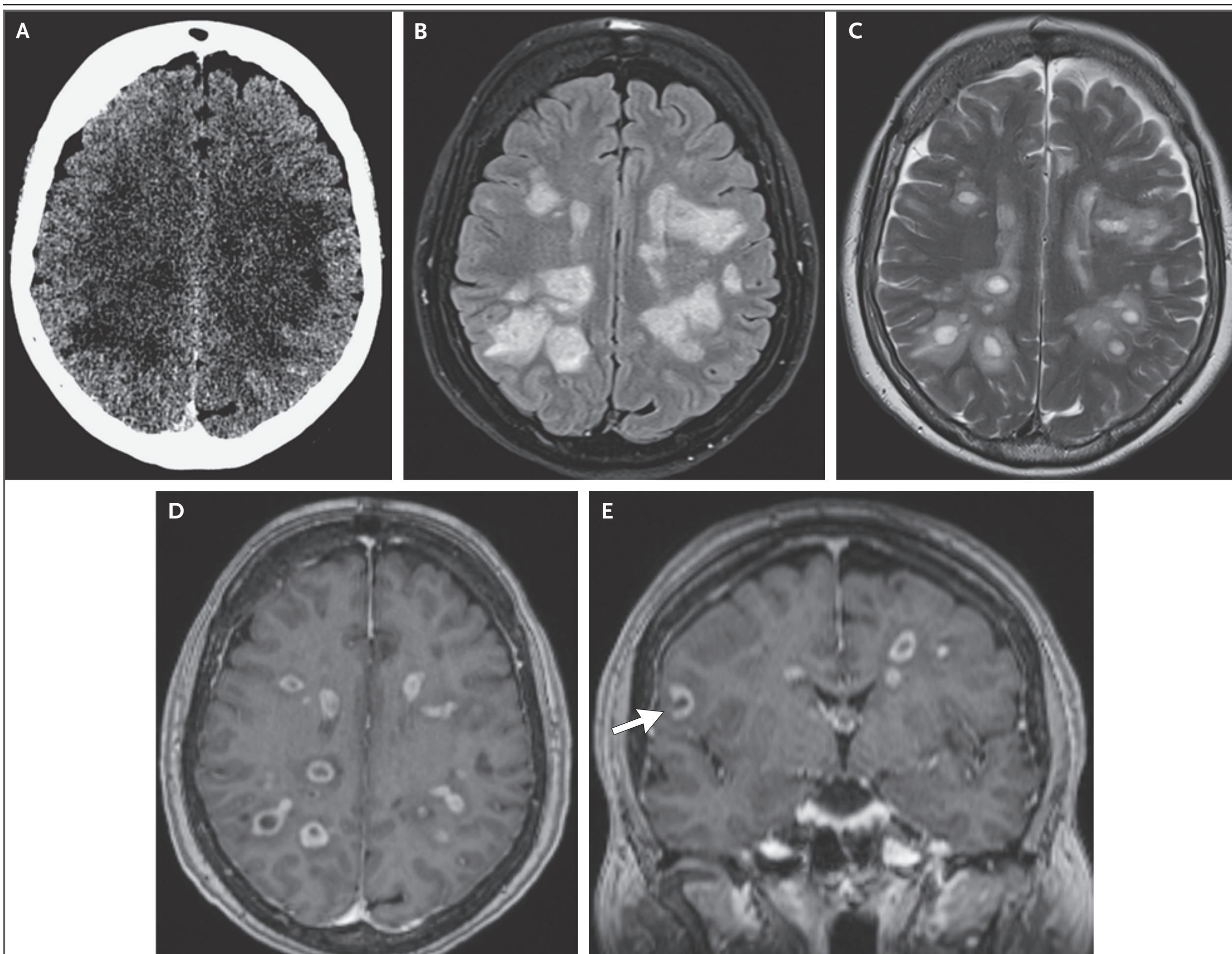


Figure 1. Initial Imaging Studies of the Head.

CT of the head was performed 4 days before the current presentation. An image obtained without the administration of contrast material (Panel A) shows scattered hypoattenuation in the cerebral white matter. MRI of the head was performed on the day of the current presentation at a magnetic field strength of 1.5 Tesla. A fluid-attenuated inversion recovery (FLAIR) image (Panel B) and a T2-weighted image (Panel C), both obtained without the administration of contrast material, show lesions within the supratentorial white matter with central signal hyperintensity, a T2-hypointense rim, and mild edema. Axial and coronal T1-weighted images (Panels D and E, respectively), obtained after the administration of contrast material, show central hypointensity with predominantly closed-ring enhancement. A subset of the lesions shows open-ring enhancement oriented toward the cortex (Panel E, arrow).

venous contrast material, showed scattered bilateral subcortical white-matter hypodensities.

Dr. Strander: Magnetic resonance imaging (MRI) of the head was recommended, but the patient elected to leave the emergency department. Her primary care physician arranged for the MRI to be performed on an outpatient basis.

Dr. Cartmell: On the day of the current presentation, outpatient MRI of the head was performed at a magnetic field strength of 1.5 Tesla with and without the administration of intravenous gadolinium-based contrast material (Fig. 1B through 1E. Numerous, mildly expansile foci of hyperintensity were seen throughout the supratentorial white matter on fluid-attenuated inversion recovery (FLAIR) images. A majority of the lesions showed central T2 and FLAIR hyperintensity with a T2-hypointense rim, a small periphery of T2- and FLAIR-hyperintense edema, central T1 hypointensity, and closed-ring enhancement on contrast-enhanced T1-weighted images; open-ring enhancement was observed in a subset of the lesions. There was no associated restricted diffusion or discernible susceptibility artifact.

Dr. Strander: The patient was instructed to present to the emergency department of this hospital for evaluation.

In the emergency department, the patient reported ongoing fatigue and feeling unsteady with walking and with climbing stairs; she had had two falls in the past several days. She also noted word-finding difficulties that were more pronounced in the evening.

Other medical history included hypertension, hypothyroidism, prediabetes, obesity, and *BRCA2*-positive status. Surgical history included appendectomy 13 years before the current presentation and prophylactic oophorectomy 5 years before the current presentation. The most recent mammogram, which had been obtained 13 months before the current presentation, showed no evidence of cancer. MRI of the breasts, which had been performed 5 months before the current presentation, showed a focus of enhancement in the right breast that was reported as being probably benign, and follow-up imaging in 6 months was recommended. Pancreatic cancer screening with CT of the abdomen and pelvis had been performed 6 months before the current presentation; no pancreatic lesions were identified, but a solid, 5-mm pulmonary nodule in the right lower lobe was noted, with follow-up imaging recommended. The patient's mother had had depression and ovarian and endometrial cancer, and a screening test performed after her mother's cancer diagnosis had been positive for a *BRCA2* variant; her paternal grandmother had had breast cancer, and her father had hypertension.

Medications included escitalopram, levothyroxine, and lisinopril–hydrochlorothiazide; lorazepam and ramelteon were used as needed for sleep, and she had recently completed a 5-day course of nitrofurantoin. She had no known adverse reactions to medications. She lived with her husband in Massachusetts and worked as an administrator in health care but had no contact with patients. She was a lifelong nonsmoker and drank alcohol occasionally; she did not use illicit drugs.

On examination, the temporal temperature was 36.9°C, the blood pressure 153/73 mm Hg, the heart rate 50 beats per minute, the respiratory rate 18 breaths per minute, and the oxygen saturation 98% while the patient was breathing ambient air. The body-mass index (the weight in kilograms divided by the square of the height in meters) was 28.1. The patient was awake, alert, and appropriately interactive and oriented. Slight inattention and slowness with stating the days of the week backward were noted. She was unable to subtract 7 from 100. Her speech was fluent without paraphasic errors. She was able to follow simple commands and to lift both arms against gravity, with slight orbiting of the left arm. Slight weakness was observed in the right leg, but she was able to walk without assistance. The remainder of the examination was normal.

The patient was admitted to the hospital, and a diagnostic test was performed.

DIFFERENTIAL DIAGNOSIS

Dr. Shamik Bhattacharyya: This 64-year-old woman presented with a 1-week history of progressive neurologic symptoms and encephalopathy. MRI of the head showed numerous hyperintensities on FLAIR images. To develop a differential diagnosis to explain the patient's symptoms, I will start by combining neurologic localization with the time course of her disease progression.

LOCALIZATION

The patient's prodromal symptoms of fatigue and poor sleep are nonlocalizing. Difficulty with concentration and with performing tasks indicates a cerebral process affecting attention. Given her impairment in attention, other cognitive findings such as making errors in calculations are less localizing. An unsteady gait, orbiting of the left arm, and weakness in the right leg do not localize to a single anatomical location. Overall, the patient's clinical findings correlate with the imaging findings of multifocal cerebral disease. Because her symptoms evolved over a period of days, the time course is considered to be subacute.

SUBACUTE MULTIFOCAL CEREBRAL DISEASE

Subacute multifocal cerebral disease can be caused by infection, cancer, rapid accrual of vascular thrombi, or immune-mediated disorders. Although I will consider possible diagnoses in each of these categories, I note the following overall observation: this patient presented with only moderate encephalopathy and neurologic dysfunction, yet a surprisingly large number of bilateral brain lesions were detected on MRI. This degree of mismatch often indicates an infiltrative process such as glioma or demyelinating disease rather than a destructive process such as ischemic stroke.

INFECTION

The patient is probably immunocompetent, given that she has no known history of human immunodeficiency virus infection, other immunosuppressive conditions, or opportunistic infections. Epidemiologic risk factors for exposure to a source of infection include travel to Maine during the summer. Causes of viral encephalitis from mosquitoes or ticks in the northeastern United States during summer include West Nile virus, eastern equine encephalitis virus, and Powassan virus. Infections with these viruses generally cause inflammation-related signal changes that can be seen on MRI of the head, rather than space-occupying lesions. Progressive multifocal leukoencephalopathy due to JC virus infection can also cause bilateral white-matter lesions but usually occurs in immunosuppressed persons without mass effect from the lesions.

Bacterial and fungal brain abscesses and parasitic infections such as neurocysticercosis and toxoplasmosis can cause space-occupying lesions in the brain. Bacterial brain abscess results from contiguous spread of head and neck infections or from hematogenous spread. Although headache is present in approximately 69% of patients with bacterial brain abscess, some affected patients may present solely with confusion and focal neurologic signs.[1] Features of brain abscess on MRI evolve from early inflammation-related changes (e.g., cerebritis) to demarcated capsular lesions (e.g., abscess). Restricted diffusion is a sensitive MRI sign that is present in more than 95% of encapsulated bacterial abscess lesions.[2] Fungal infections can cause abscesses that have irregular walls with varied cavitary signal changes, but portions of the abscesses typically show restricted diffusion on MRI.[3] Given the absence of restricted diffusion in all of the enhancing brain lesions in this patient, the presence of bacterial or fungal abscess is unlikely.

The patient does not have known exposure to cats that would suggest an increased risk of toxoplasmosis, nor does she have severe immunocompromise that would suggest an increased risk of reactivation of toxoplasmosis leading to brain lesions. In addition, the lesions detected on MRI did not have the targetoid appearance that is often present in lesions associated with toxoplasmosis.[4] Cysticercosis has been reported throughout the United States, but the highest concentration of severe disease is in the western United States.[5] Seizure is more likely to manifest with neurocysticercosis than with systemic cysticercosis, and imaging typically shows a single cysticercal lesion or a small number of these lesions; however, multiple lesions can be present.[6] MRI findings vary depending on the stage of the cyst, but in high-resolution images, the scolex of *Taenia solium* (pork tapeworm) often appears as an eccentric nodule — a finding that was not observed in this patient. Overall, toxoplasmosis and cysticercosis are possible but unlikely diagnoses in this patient.

CANCER

Both primary cancer of the central nervous system (CNS) and metastatic disease involving the CNS can cause subacute confusion and imbalance. This patient has a known *BRCA2* risk allele, which confers a lifetime risk of breast cancer of approximately 70%.[7] In addition, a pulmonary nodule in the right lower lobe was detected on imaging 6 months before the current presentation. The presence of bilateral multifocal brain lesions in this patient is suggestive of metastatic disease, which may arise from breast, lung, kidney, skin, colorectal, or other cancers.

The patient's relatively preserved neurologic function in the context of numerous brain lesions, the absence of clustering of lesions at the gray–white junction, and only minimal perilesional edema would be unusual for brain metastases.[8,9] Lymphoma can also cause multifocal brain masses, but in immunocompetent adults, imaging findings are typically characterized by homogeneously gadolinium-enhancing lesions with associated restricted diffusion.[10]

VASCULAR THROMBOSIS

Thrombotic thrombocytopenic purpura, antiphospholipid syndrome, infective endocarditis, and vasculitis can all cause rapid accrual of small ischemic lesions, leading to subacute ischemic encephalopathy. Some of these disorders can also result in systemic thrombotic disease, such as splenic or renal infarction, which this patient does not have. Infective endocarditis would be unlikely in this patient, given that she had no fever, leukocytosis, or elevated levels of inflammatory markers. In patients with a vasculitic disease, MRI of the head typically shows ischemic infarcts at different stages of evolution, findings that were not observed in this patient. Primary CNS vasculitis may cause masslike lesions alone in rare cases and remains a possible diagnosis in this case.

IMMUNE-MEDIATED DISORDER

Immune-mediated disorders that cause focal brain lesions can be grouped broadly into demyelinating disorders, systemic autoimmune diseases with neurologic involvement, and primary autoimmune encephalitides. Multiple sclerosis (MS) is the most common demyelinating disease, with an average age at diagnosis of 32 years.[11] Over the past several decades, however, the number of persons older than 50 years of age in whom MS is diagnosed has increased, and this age group now accounts for approximately 10% of new diagnoses.[12] Therefore, age alone does not rule out MS.

Lesions associated with MS are found in the juxtacortical, periventricular, and cerebellar regions and in the brain stem and spinal cord. On contrast-enhanced imaging in this patient, some lesions had an open-ring configuration, which is a radiologic sign that distinguishes demyelinating lesions from tumors and infections.[13] With open-ring enhancement, the closed portion of the ring abuts the white matter, and the ring faces the ependymal surface. In meta-analyses, the pooled incidence of the open-ring sign was 35%, but the pooled specificity for distinguishing demyelinating lesions from brain tumors was more than 95%.[14] Limited perilesional edema is also a characteristic feature of demyelinating disease. This patient has a multitude of lesions with only moderate clinical symptoms and MRI features including open-ring enhancement and limited perilesional edema, findings consistent with a diagnosis of late-onset MS. Lesions seen on MRI typically appear as small, circumscribed, multifocal lesions or as larger, masslike lesions (also referred to as tumefactive demyelinating lesions).

Two other important causes of demyelination are aquaporin-4 (AQP4) antibodies and myelin oligodendrocyte glycoprotein (MOG) antibodies, and testing for these should be performed. This patient has no clinical history of systemic autoimmunity, but some rheumatologic disorders, especially sarcoidosis, may manifest as CNS disease initially. MRI findings in persons with neurosarcoidosis typically show leptomeningeal gadolinium enhancement, which was not seen on MRI in this patient.[15]

Primary autoimmune encephalitides are typically characterized by inflammatory changes rather than the discrete, masslike enhancing lesions that were observed on MRI in this patient. In addition, the absence of blood products on MRI in this patient would be unusual for inflammation associated with cerebral amyloid angiopathy.

DIAGNOSTIC STEPS

Although my leading diagnosis is late-onset MS, the information provided does not fully establish this diagnosis. Additional testing is necessary, including cross-sectional body imaging, evaluation of cerebrospinal fluid (CSF), and serologic testing for other demyelinating disorders. If a diagnosis is not established after such testing, a brain biopsy may be indicated. Diagnostic brain biopsy performed in the context of an unclear neuroinflammatory or oncologic disorder leads to a diagnosis in approximately 60% of patients and results in a change in treatment in approximately 77% of patients; 6% of patients have complications from the biopsy for which they receive treatment.[16]

CLINICAL IMPRESSION AND INITIAL MANAGEMENT

Dr. Albert Y. Hung: Given the patient's clinical presentation and imaging findings, we focused the differential diagnosis on infection, demyelinating processes, and cancer. Infection was unlikely, since the patient did not report fever, chills, or headache, and laboratory studies did not show leukocytosis or elevated levels of inflammatory markers. To evaluate for the possibility of cancer with CNS metastases, we performed CT of the chest, abdomen, and pelvis, which showed a 5-mm lung nodule in the right lower lobe that was unchanged from that seen on a study performed 6 months earlier; subcentimeter left supraclavicular lymph nodes were also noted.

Lumbar puncture was performed, with an opening pressure of 21 cm of water. Analysis of the CSF showed mild lymphocytic pleocytosis, with 6 to 9 nucleated cells per microliter (reference range, 0 to 5), of which 95% were lymphocytes. The CSF glucose level was 62 mg per deciliter (3.4 mmol per liter; reference range, 50 to 75 mg per deciliter [2.8 to 4.2 mmol per liter]), and the total protein level was 57 mg per deciliter (reference range, 5 to 55). No organisms were seen on Gram's staining, and testing for toxoplasma IgM and IgG was negative. Cytologic evaluation of the CSF showed no malignant cells, and a nucleic acid amplification test for a *MYD88* mutation was negative. A CSF sample was sent for testing for the presence of oligoclonal bands and measurement of the level of free kappa light chain.

Whole-body positron-emission tomography (PET) did not show ^{18}F-fluorodeoxyglucose (FDG)–avid lesions, although an FDG-PET scan of the head showed multifocal uptake in the supratentorial white-matter abnormalities that had previously been noted on MRI. MRI of the cervical and thoracic spine did not show any spinal cord abnormalities. Blood tests for MOG and AQP4 antibodies were negative.

On the basis of these results, we thought that an inflammatory or demyelinating disease was the most likely diagnosis in this patient. However, she did not have any preceding illness that would suggest postinfectious acute disseminated encephalomyelitis, and her age would be atypical for an initial presentation of a primary demyelinating disease. The neuro-oncology consultant agreed that a primary CNS cancer could not be ruled out. After a multidisciplinary discussion, and in the absence of a clear diagnosis, the neurosurgery team performed a brain biopsy for definitive tissue diagnosis.

CLINICAL DIAGNOSIS

Inflammatory or demyelinating disease.

DR. SHAMIK BHATTACHARYYA'S DIAGNOSIS

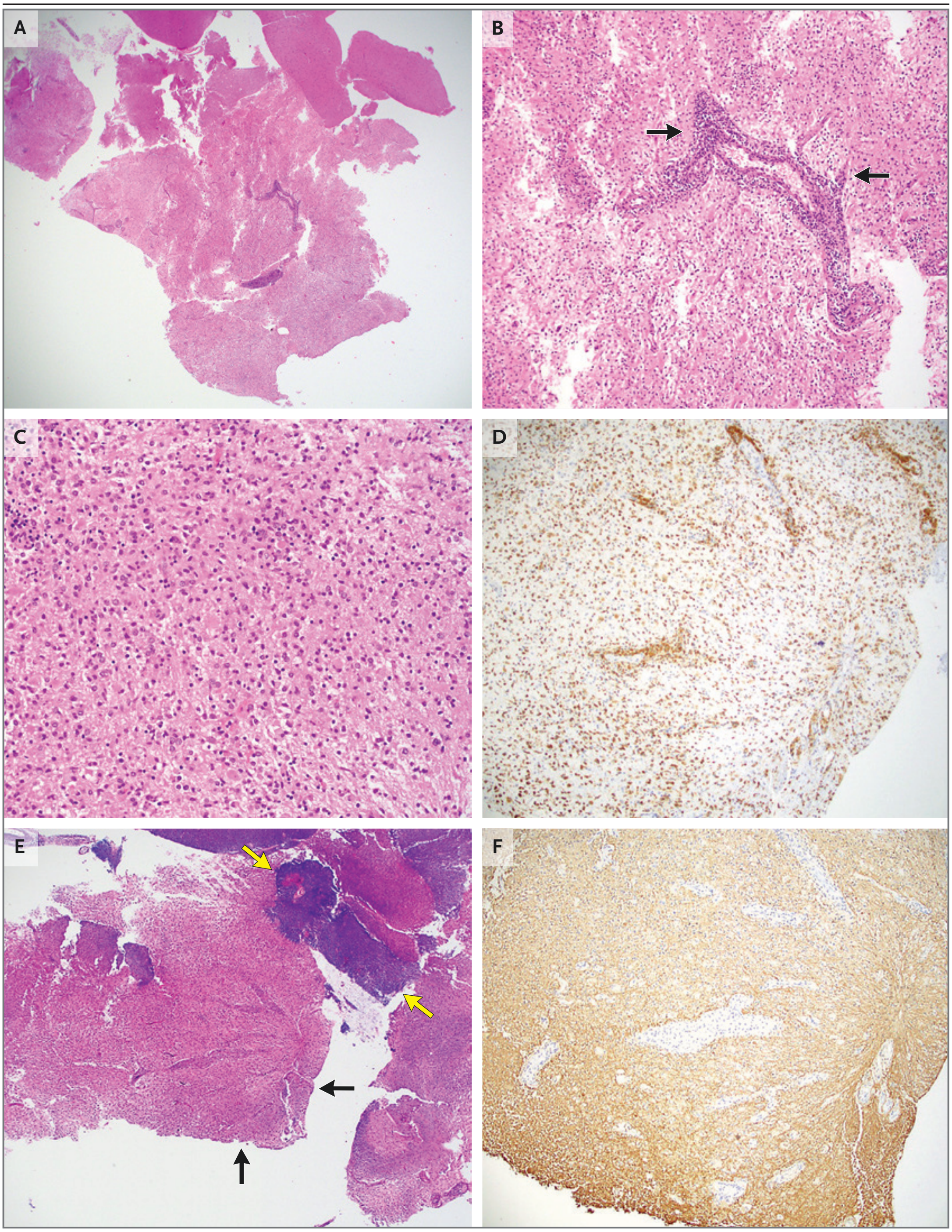
Late-onset multiple sclerosis.

DIAGNOSTIC TESTING

Dr. Elizabeth C. Conner: Hematoxylin and eosin staining of sections obtained from multiple brain lesions in the right frontal lobe (Fig. 2A, 2B, and 2C) showed parenchyma with a cellular lesion consisting of a mixed inflammatory infiltrate with abundant macrophages admixed with small, mature lymphocytes. No granulomas or viral cytopathic effects were observed, and all the biopsy specimens had a similar histologic

Figure 2 (facing page). Biopsy Specimens of the Brain.
Hematoxylin and eosin staining of sections obtained from multiple lesions in the right frontal lobe show brain parenchyma involved by a cellular, mixed inflammatory lesion consisting of numerous plump, eosinophilic macrophages (Panels A, B, and C) and parenchymal and perivascular lymphocytes (best seen in Panel B, arrows). Immunohistochemical staining for CD68 (Panel D) shows sheets of macrophages within the lesion. Luxol fast blue–hematoxylin and eosin staining (Panel E) shows loss of myelin within the lesion, which appears pink (black arrows) in contrast to the adjacent blue, myelinated white matter (yellow arrows). Immunohistochemical staining for neurofilament (Panel F) shows relative preservation of axons within the white matter of the lesion. Macrophage-rich inflammatory lesions with loss of myelin and preservation of axons are consistent with acute demyelinating lesions. Additional testing for infectious sources (not shown) was negative.

appearance. Immunohistochemical staining for CD68 (Fig. 2D) confirmed the presence of sheets of macrophages within the lesion and surrounding vasculature. Luxol fast blue–hematoxylin and eosin staining (Fig. 2E) showed distinct loss of myelin, whereas immunohistochemical staining for neurofilament (Fig. 2F) showed relative preservation of axons, which is characteristic of an acute demyelinating lesion. On additional immunohistochemical staining, the lymphocytic infiltrate was composed primarily of CD3+ T lymphocytes that involved the lesion and surrounding



vessels, with a smaller population of CD20+ B lymphocytes, which were confined to the perivascular cuff — findings that are consistent with a reactive inflammatory infiltrate with no evidence of cancer. Although perivascular inflammation was present, there was no evidence of involvement of the vessel walls or vasculitis.

With histologic confirmation of demyelinating lesions, it is imperative to investigate possible infectious causes, particularly infection with JC virus. Staining for simian virus 40 antigen (a T antigen with reactivity to JC virus) was negative, which ruled out progressive multifocal encephalopathy. Additional staining was also negative, including Grocott methenamine–silver staining for infectious organisms, immunostaining for varicella–zoster virus and spirochetes, and in situ hybridization for Epstein–Barr virus–encoded small RNA.

The overall findings — lesions containing numerous macrophages, with loss of myelinated fibers and preservation of axons — are histologically consistent with acute demyelinating lesions. In the absence of an infectious cause, identification of demyelinating lesions prompts consideration of a number of inflammatory and autoimmune conditions, including MS, acute disseminated encephalomyelitis, neuromyelitis optica spectrum disorder (NMOSD), and MOG antibody–associated disease (MOGAD), which are characterized by different clinical, serologic, and radiologic features.[17]

PATHOLOGICAL DIAGNOSIS

Demyelinating disease.

DISCUSSION OF MANAGEMENT

Dr. Philippe A. Bilodeau: After the results of the brain biopsy became available, the unifying diagnosis in this patient was tumefactive demyelination, defined by the presence of at least one demyelinating lesion measuring at least 2 cm in diameter. Tumefactive demyelination is a heterogeneous syndrome that can be characterized by a variety of clinical and radiologic findings rather than a single disease entity. In a retrospective cohort involving more than 250 patients with tumefactive demyelination, 70% had tumefactive MS, 5% had MOGAD, and 24% had tumefactive demyelination that remained cryptogenic.[18]

First-line treatment for tumefactive demyelination consists of high-dose intravenous methylprednisolone for 3 to 5 days. This patient's symptoms abated after treatment with glucocorticoids. If she had had clinically significant disability, plasma exchange would have been considered. In cases of life-threatening cerebral edema, treatment with intravenous cyclophosphamide is an option, and there is emerging evidence for the use of intravenous tocilizumab in select cases.[19]

The initial evaluation of patients presenting with tumefactive demyelination focuses on distinguishing MS from other inflammatory demyelinating disorders, such as MOGAD and NMOSD, and from noninflammatory causes of white-matter lesions, including infections and cancer. Testing for MOG IgG and AQP4 antibodies was negative, and histologic examination of brain-biopsy specimens showed no evidence of cancer or infection. The level of free kappa light chains was elevated, and CSF analysis revealed 8 CSF-restricted oligoclonal bands (i.e., bands were present in the CSF but not in the blood) (reference value, <2). Additional imaging was performed.

Dr. Cartmell: A vein coursing through the central portion of a lesion (the central vein sign) and paramagnetic signal along the lesion rim (paramagnetic rim lesions) are specific imaging features of MS. Anatomical features of the veins and paramagnetic signal are best visualized on T2*-weighted sequences on a 7-Tesla MRI scanner. Approximately 7 weeks after the current presentation, MRI of the head was performed with a 7-Tesla machine before and after the administration of intravenous contrast material. Numerous FLAIR-hyperintense lesions were again seen in the supratentorial white matter, but contrast enhancement had diminished and edema had decreased. When T2*-weighted and FLAIR images were overlaid, a central vein was visible within many of the deep and periventricular lesions (Fig. 3). Paramagnetic rim lesions were not observed.

Dr. Bilodeau: Together, these findings supported a diagnosis of MS on the basis of the 2024 McDonald criteria.[20] Initiation of long-term, disease-modifying therapy is warranted in all patients with tumefactive MS. Given the aggressive

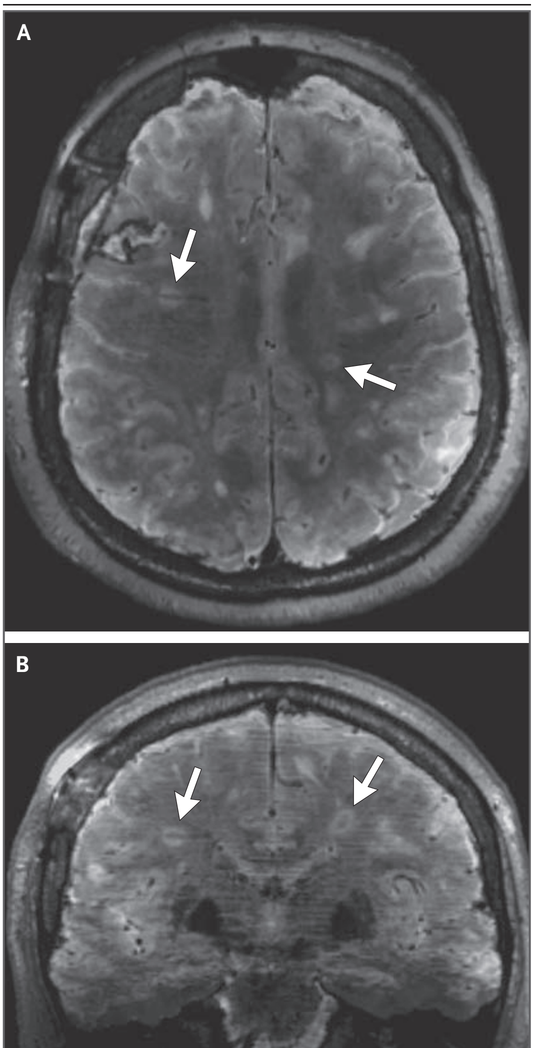


Figure 3. Follow-up MRI of the Head.

MRI of the head was performed approximately 7 weeks after the current presentation. Axial and coronal FLAIR–T2*-weighted fusion images (Panels A and B, respectively) show that the lesions have slightly contracted in size as compared with that observed on initial imaging, and a central vein is visible in a majority of the lesions in the deep and periventricular white matter (arrows). A rim of paramagnetic susceptibility was not observed.

nature of the disease, highly efficacious disease-modifying therapy such as B-cell–depleting agents or alemtuzumab is typically preferred. However, this patient's older age at presentation added therapeutic complexity.

Approximately 1% of MS cases are diagnosed in patients older than 60 years of age.[12] Late-onset MS is more likely to occur in women than in men and is associated with more extensive motor involvement and a higher risk of progressive disease than MS with onset at an earlier age.[21] Although the efficacy of disease-modifying therapy declines with age, subgroup analyses performed in clinical trials and observational studies have consistently shown a greater benefit with highly effective therapies than with agents with low or moderate efficacy, even among older adults.[22,23]

Although the use of highly effective agents is still preferred in older adults, these patients are at increased risk for infection, probably owing to immunosenescence.[24] Infections are an important consideration, since nosocomial infections can worsen disability in persons with MS.[25] In addition, coexisting conditions are more common in this age group and may further influence selection of disease-modifying therapy.

After discussion of risks and benefits, which took into account the patient's age, the risk of infection, and the aggressiveness of disease, treatment with ublituximab — a CD20 monoclonal antibody — was initiated. One month later, her cognition had normalized, although her severe fatigue persisted. She had no new focal neurologic symptoms. Neurologic examination was largely normal, with the exception of an impaired tandem gait. The Expanded Disability Status Scale score was 1.0 (on a scale of 0 to 10.0, with higher scores indicating more severe disease).[26] A previous score was not available for comparison.

Final Diagnosis

Multiple sclerosis.

This case was presented at the combined Neuroscience Institute–Neurology Grand Rounds.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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